

# Rohan Shukla

4<sup>th</sup> Year Major in Computer Science

[LinkedIn](#) | [Email](#) | [GitHub](#) | [Phone](#)

## Technical Skills

**Programming Languages:** C++, Java, Python, JavaScript, C, Assembly, R, HTML, CSS, TypeScript, SQL

**Tools:** VS Code, IntelliJ, RStudio, GDB, Git/GitHub, JSON, TensorFlow, ReactJS, Oracle, RoboFlow, Django

## Work Experience

**Software Engineer** | *Nexera Robotics*

**May 2025- Dec 2025**

- Refactored a 10-module threaded pipeline using Ray, eliminating redundant compute and fixing task/worker misconfigurations. Reduced overall cycle time by 47% (15s → <8s) and robot dwell time to zero.
- Re-architected a 25+ table PostgreSQL schema handling >1,000 inserts/7s: restored normalisation, established consistent migration patterns, and improved query efficiency for Grafana and Power BI.
- Automated 3D camera calibration using a 2D detection model, point cloud projection, and surface matching — eliminating a daily manual process and improving bin pose accuracy for downstream collision detection and pose ranking.
- Deployed a material segmentation model (>95% accuracy) into a production robotics pipeline, directly contributing to a >96% pick success rate.

**Machine Learning Engineer** | *UBC Agrobot*

**Sep 2024- Sep 2025**

- Training Faster R-CNN model for bounding box detection using Python and ARC Sockeye achieving 95% accuracy
- Implementing a real-time video feed processing system capable of analyzing 30 FPS, enabling sequential frame analysis with minimal latency.
- Developed scripts to parse and convert various dataset annotations into YOLO format

## Technical Projects

**Live Personal Trainer** | *Python*

**Nov 2024 – Jan 2025**

- Integrated MediaPipe's **BlazePose machine learning model** for real-time human pose detection, tracking key body landmarks such as joints and limbs from live video feeds
- Worked closely with fitness professionals to develop a comprehensive set of exercise form guidelines
- Designed a system to overlay live feedback on the user's video feed, highlighting key areas where posture deviates from optimal form

**Insight Façade** | *TypeScript, ReactJS, Mocha, JSON*

**Sep 2024 – Dec 2024**

- Collaborated with a team to develop a **full stack React** application managing 5000+ class records, allowing users to input store and search databases with advanced filtering
- Built a comprehensive test suite using Mocha and Chai, achieving 95% test coverage for core functionalities
- Refactored code to optimize the dataset addition process resulting in a 75% increase in performance
- Utilized async promises to parse and verify large datasets and store them on disk as JSON files

**Discord AI Moderator/Assistant** | *Python, OpenAI API, Discord*

**Jul 2024 – Aug 2024**

- Developed a Discord bot deployed on 100+ servers to monitor chat activity and enforce server guidelines
- Leveraged Python scripting with GPT-4 and Discord API to create a bot that assists users with natural language interactions and detects inappropriate language, automatically enforcing rules
- Maintained session history of over 5,000 users across all hosted servers, enabling the bot to retain context and provide consistent, context-aware responses in all sessions

## Education

**University of British Columbia**

**Expected May 2027**

**Bachelor of Science – Computer Science**

**CGPA: 4.0**